# Intelligent Verification System for Concrete Compliance & QA Risk Management

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# Section 1: Company Overview

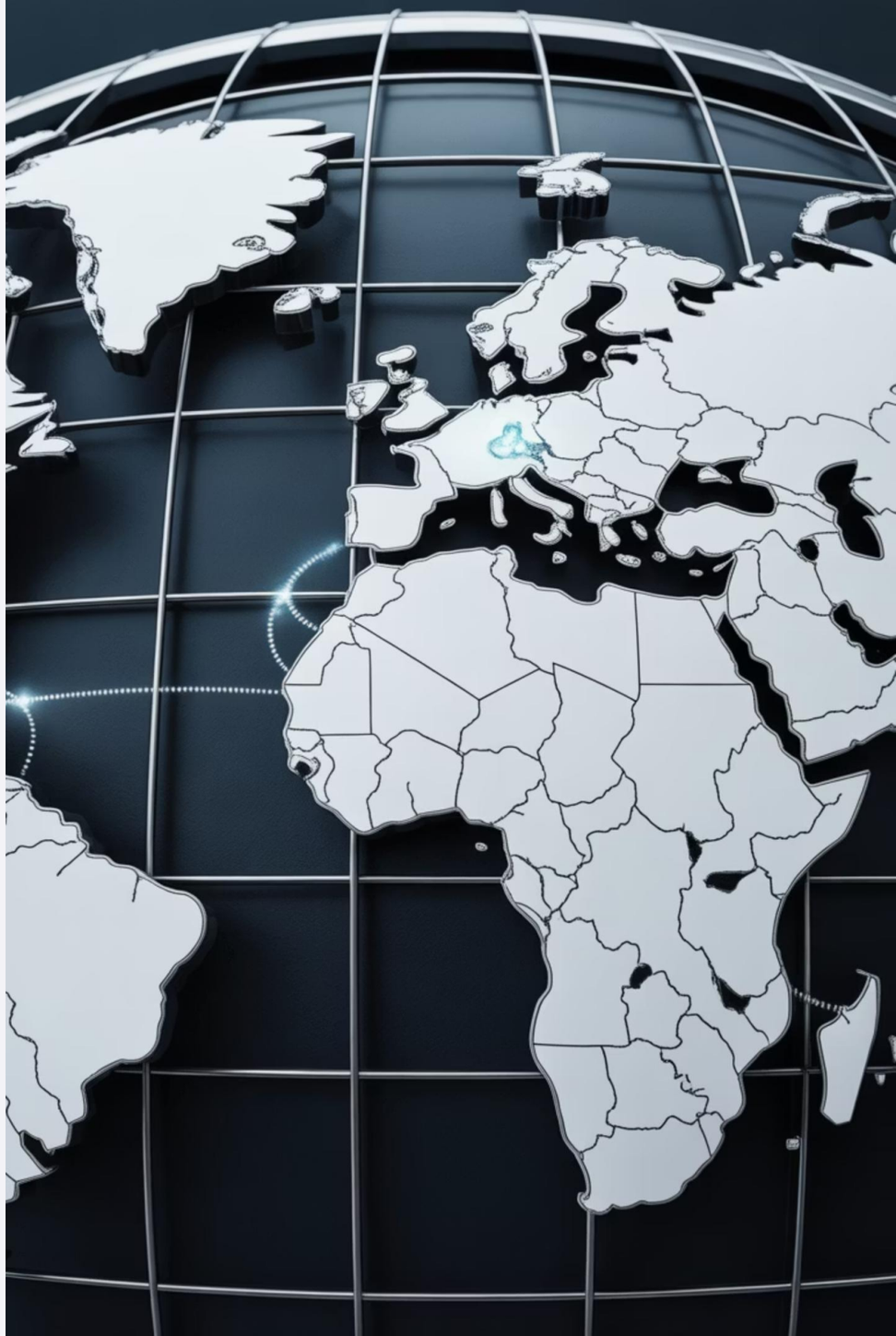
# Company Overview

**Ambient Systems** is an AI-driven compliance and sustainability venture re-establishing operations in Singapore to conduct a BEAMP collaboration. We combine cutting-edge artificial intelligence with deep domain expertise to transform how the built environment manages quality assurance and regulatory compliance.

Spun-off from the National University of Singapore in 2021, Ambient Systems rapidly expanded its with operations spanning **Serbia (HQ), Hong Kong**, and **Quebec**. Our technology has been validated across multiple international markets and recognized by leading innovation platforms.

## Recognition & Innovation

- **WIPO Patent WO –** Proprietary AI framework
- **Panasonic Deep Tech Challenge Winner** (2023)
- **Google Runner-up** (2022)



# Business Activities

**Ambient Systems** develops **AI platforms that connect financial, operational, and sustainability data** to deliver measurable, auditable performance improvements across industries such as *construction*, *infrastructure*, *energy*, and *supply chain systems*.

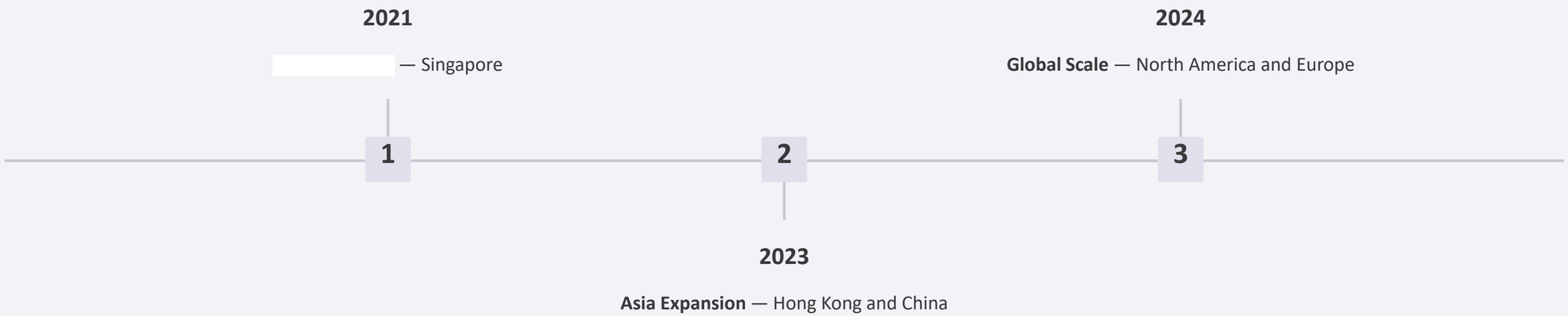
### Platform

- Financial Metrics
- Generative AI Planning
- IoT Connectivity

### Consulting

- Civil Engineering & Energy Systems
- Controlled Environment Agriculture
- Intellectual Property and R&D

## Global Market Presence




Singapore serves as our strategic testing and validation hub, enabling us to build robust solutions before scaling globally.



# Products / Services

- Ambient Systems offers three integrated offerings specifically designed to address the challenges outlined in the BEAMP Cycle 6 program.
- Each solution builds upon our core AI and compliance technology platform.

# 1. POC / Pilot — BEAMP Cycle 6 Focus

Our pilot solution aims to deliver a focused AI-driven verification that addresses critical pain points in construction quality assurance and requirements.

### Concrete Quality

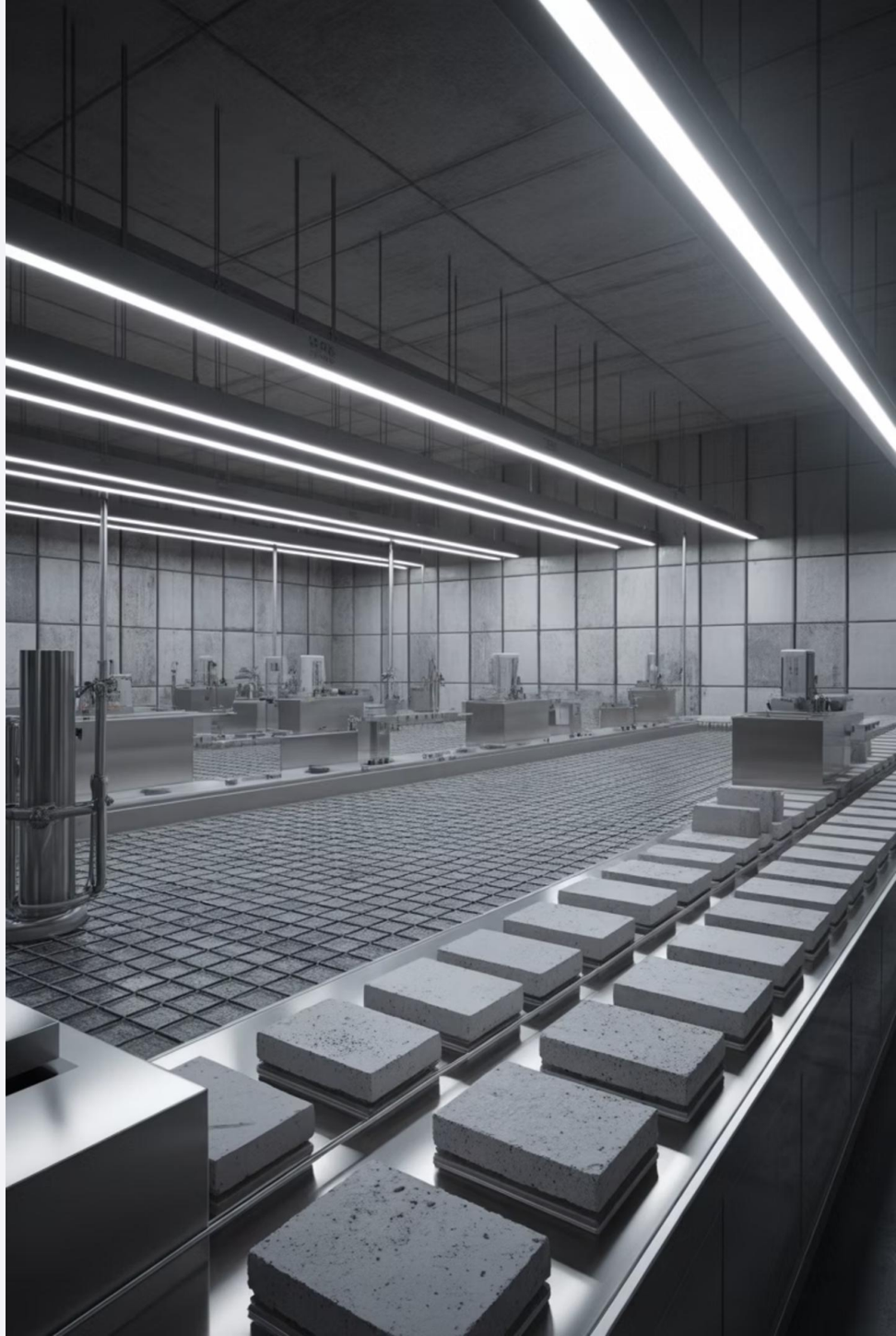
Automated verification of delivery orders, slump tests, and cube-test data. Proposed solution, discussed in the next section, ingests data from multiple sources — WhatsApp logs, PDFs, spreadsheets — and reconciles them into a unified permanent record.

### Construction Economics

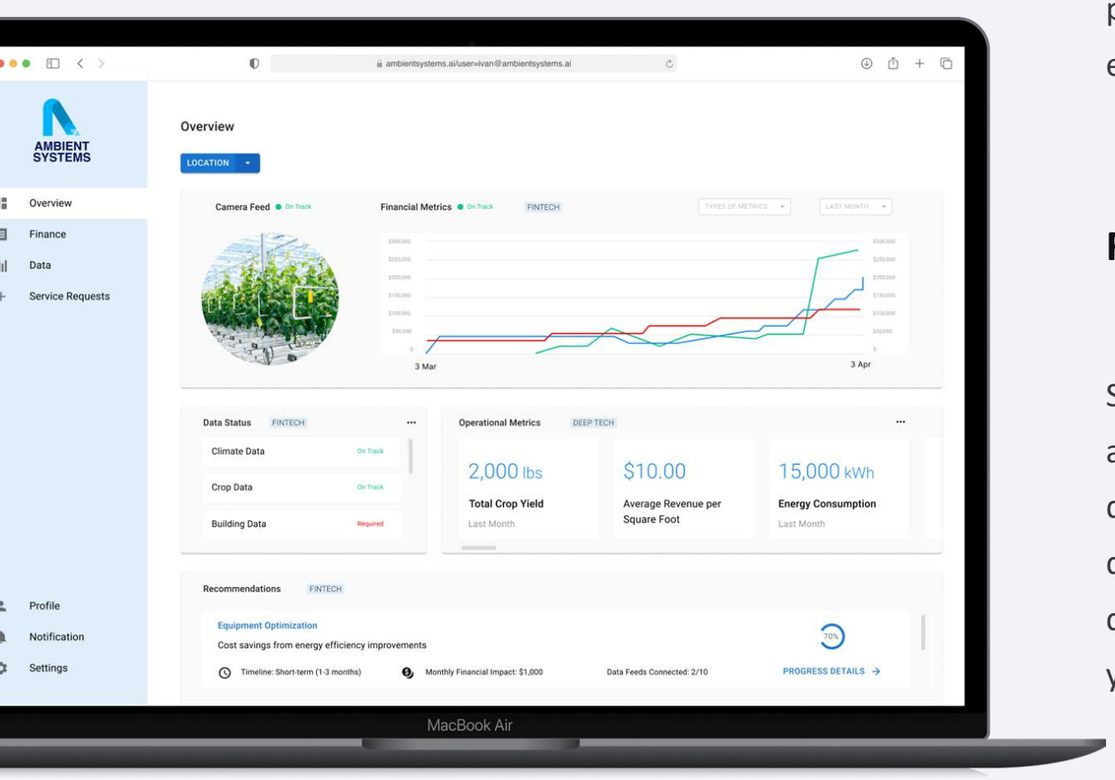
Intelligent linking of QA performance to financial metrics and project milestones. Track how quality compliance impacts cost, schedule, and payment approval workflows in real-time.

### Sustainability Metrics

Comprehensive tracking of embodied carbon and materials data for ESG reporting. Transform QA data into sustainability insights that meet Singapore's Green Building standards and international ESG frameworks.



# 2. Platform (ambientsystems.ai)




Ambient Systems platform comprehensively integrates quality assurance, financial performance, and sustainability analytics into a unified intelligence system. Built on cutting-edge AI and designed for seamless integration with existing construction technology stacks.

### Financial Metrics

Seamlessly integrates QA and project performance data with cost visibility dashboards. Track how quality compliance impacts your bottom line.

scenario modeling for compliance forecasting and construction logistics optimization. Predict potential issues before they occur.

### IoT Connectivity

Direct interfaces with sensors and delivery data systems enable automated real-time verification without manual data entry.

The platform serves as the central nervous system for construction compliance, connecting disparate data sources and transforming them into actionable intelligence for project teams, auditors, and stakeholders.

# 3. Consulting (www.ivandamnjanovic.com)

Beyond technology, we provide deep technical consulting services that leverage our founder's extensive expertise across multiple engineering domains and intellectual property development.



## Civil Engineering

Structural QA, performance audits, and digital twin modeling for infrastructure projects. We bring rigorous engineering analysis to compliance verification.



## Energy Systems

Renewable energy integration, load optimization, and carbon analytics. Helping projects achieve net-zero targets through data-driven strategies.



## Controlled Environment Agriculture

Data-driven optimization of indoor production systems, applying our AI expertise to sustainable food production.



## Intellectual Property

Systems design and patented AI compliance frameworks (WO ). We help organizations protect and monetize their innovations.

# Business Targets

## 1 – 3 Year

### Client Tester Acquisition

Secure LOIs/MOUs and advance Technology Readiness Level from 6 to 9

### Intellectual Property Establishment

Secure Singapore-based IP portfolio and regulatory approvals

### Complete Platform Development

Finalize core modules and achieve production-ready status

## Singapore Entity Strategy

**Ambient Systems** is establishing a dedicated Singapore business focused on **applied AI and computing, sustainability, and enterprise performance verification** in the built environment – among other adjacent sectors. This new entity will leverage our existing R&D and IP portfolio (including patent WO ) and expertise while serving as a new venture designed to co-develop with BEAMP and additional partners.

## Global Validation

Our current operations across **Canada, Hong Kong, and Serbia** have validated the urgent market need for integrated ESG-AI systems in multiple regions. The **BEAMP programme** offers the ideal opportunity to anchor these proven capabilities in Singapore, then scale solutions to clients across the globe.

# Key Team Members

**Ivan Damnjanovic**
**Founder & Systems Architect**

- **Entrepreneur / Inventor** from Belgrade with almost 10 years of experience across Asia (Singapore, Nepal, Hong Kong).
- Track record scaling YC-backed startups and cross-border ventures
- *Languages: Native Serbian & English (fluent); Intermediate French*

**Advertisement & Customer Success Specialist**

- Advertising specialist at **Yourbow.com**, bringing internet-first commercial advertising experience to diverse audiences.
- Based in Belgrade, Serbia.
- Experienced with diverse stakeholder communications in online and tourism industries.



# Section 2: Proposal

# Proposal Summary

**Challenge Statement (CS 9):** Intelligent Verification System for Concrete Compliance and QA Risk Management

**Challenge Statement Owner (CSO):** Expand Construction Pte. Ltd.

## Our Solution

Ambient Systems proposes an **AI-driven verification platform** that automates data ingestion, reconciliation, and compliance verification across diverse document formats — PDFs, WhatsApp logs, spreadsheets, and e-delivery orders. The system unifies QA/QC, finance, and sustainability datasets into a single dashboard for real-time traceability and ESG-linked performance assurance.

### Technology Core

Unified AI engine combining **OCR, NLP, and machine learning** for automated QA verification. Reconciliation logic linking Delivery → Slump → Cube → Approval workflows. Role-based dashboards delivering instant anomaly detection and ESG metrics.

- Real-time traceability & audit readiness
- Structured QA datasets for compliance & ESG analytics
- Intelligent risk alerts for claims and approvals

### End-to-End Coverage

From delivery orders through testing to audit readiness, with financial and compliance considerations integrated at every stage of the process.

# Introduction

## The Problem Context

Concrete QA data today is **fragmented across WhatsApp, Excel, eDO portals, and emails** — making reconciliation slow, error-prone, and unverifiable at audit time. Expand Construction currently manages **100–200 delivery orders per pour**, each requiring manual cross-checks between DOs, cube tests, and invoices. This creates significant operational inefficiency and compliance risk.

## Solution Functions & Features

1

### AI/NLP Engine

Extract key fields from semi-structured documents automatically

2

### Reconciliation Logic

Link DO → Slump → Cube → Approval with intelligent matching

3

### Role-Based Dashboards

Tailored interfaces for QA, Finance, and Audit teams

4

### Configurable Alerts

Proactive detection of missing or non-compliant records

5

### Seamless Integration

Works with Autodesk Construction Cloud & supplier portals

### Value Proposition

- Reduces QA collation time by up to **70%**
- Ensures **100% traceability** across workflows
- Supports Singapore's Construction 4.0 and ESG standards

### Competitive Advantages

- Engineering expertise + explainable AI
- Patented IP (WO )
- Ambient Verify + Ambient Twin integration

### Integration Philosophy

Designed to **enhance existing tools** — not replace them. Works alongside your current technology stack.

# Technology Architecture

Our solution is built on a sophisticated multi-layer architecture that combines cutting-edge AI with robust engineering principles. Each component is designed for scalability, security, and seamless integration with existing construction technology ecosystems.

### AI / NLP Ingestion Engine

Automated parsing of WhatsApp, PDF, Excel, and e-DO data; extracts key fields (DO, slump, cube ID, strength)

*Tech Stack: OCR + NLP (LayoutLM / GPT) · Python · Pandas*

### Reconciliation Logic

Links DO → Slump → Cube → Approval; detects mismatches and anomalies with intelligent pattern recognition

*Tech Stack: SQL + Rule-Based Engine + ML Consistency Model*

### Alert & Workflow System

Real-time detection of missing data or non-compliance; triggers notifications to relevant stakeholders automatically

*Tech Stack: Event Triggers · Threshold Rules · Automation Scripts*

### Role-Based Dashboards

Interactive visualizations for QA, Finance, and Audit teams; exportable audit reports with full traceability

*Tech Stack: React / Streamlit · Supabase / PostgreSQL*

### ESG Analytics Layer

$CO_2$ and energy traceability; sustainability & carbon-linked KPIs powered by Ambient Twin

*Tech Stack: Ambient API · Carbon Factor Database · ESG Models*

### Security & Compliance

Access control, encryption, data audit logs; CSA-aligned data protection standards

Tech Stack: OAuth 2.0 · AES-256 · CSA Cyber Essentials (SG GovTech)

# POC / Pilot Plan

## Pilot Overview

- **Objective:** Pilot AI-based verification on active project (NS Square)
- **Duration:** 4-6 months of core development + 2 months validation
- **Budget:** SGD 40,000 – 50,000
- **Team:** 2 Data Scientists · 1 Systems Engineer · 1 QA Specialist

## Stakeholders & Roles

**Ambient Systems:** AI system design, implementation, and technical support

**Expand Construction:** Data access, site validation, feedback loop, and construction management ex

**BEAMP Ecosystem:** Mentorship, funding support, and network connections

1

**Phase 1 — Month 1**

Data mapping & sample collection (WhatsApp, PDFs, Excel). Establish data pipelines and validation protocols.

2

**Phase 2 — Month 2-3**

AI ingestion prototype development + automated data reconciliation engine build-out.

3

**Phase 3 — Months 4-5**

Dashboard deployment & live testing with real project data from NS Square.

4

**Phase 4 — Month 6-7**

Validation & ROI assessment. Measure time savings and accuracy improvements.

5

**Phase 5 — Month 8-9**

Integration into Expand's ecosystem. Handover and training for full adoption.

# Impact

## Quantitative Benefits

**70%**

### Time Reduction

Reduction in manual QA collation time — from days to hours

**2**

### Days Saved

Working days saved per month per QA team through automation

**100%**

### Traceability

Complete traceability for audit compliance and claims management

## Qualitative Impact

### ESG Excellence

Real-time ESG and $CO_2$ tracking via Ambient Twin layer enables comprehensive sustainability reporting aligned with Singapore's Green Building standards.

### Financial Confidence

Greater confidence in milestone payments and audit readiness reduces financial risk and accelerates project cash flow.

### Staff Empowerment

Empowers staff with "AI personal assistants" that handle repetitive tasks, allowing them to focus on high-value strategic work.

### Manual Process

- Fragmented data across multiple platforms
- Hours of manual reconciliation per pour
- High error rates and missing records
- Delayed audit preparation

### Intelligent QA Verification

- Unified data dashboard with real-time updates
- Automated reconciliation in minutes
- 99%+ accuracy with anomaly detection
- Audit-ready reports on demand

# Scalability

Our vision extends far beyond the initial pilot. We're building a scalable platform architecture that can expand across geographies, sectors, and compliance domains to become the industry standard for intelligent verification.

### Singapore Entity

New Ambient Systems Singapore venture focused on AI operational verification for the built environment. Anchor operations and establish regional hub.

### Global Scaling

Deploy in Hong Kong and Canada for cross-jurisdiction testing and commercial proof. Validate technology across diverse regulatory environments.

### Sector Expansion

Extend to steel verification, prefabrication QA, and BIM compliance modules. Create comprehensive construction verification suite.

### Long-Term Vision

Create an **interoperable digital twin for enhanced financial performance** -- a unified QA and ESG verification layer for Construction 4.0. This will serve as the intelligent infrastructure enabling the next generation of smart, sustainable, and compliant building practices globally.

## Singapore Hub

R&D and BEAMP pilot

## Hong Kong

Asia-Pacific expansion



## Quebec

North American base

## Serbia

Europe operations

# Intelligent Verification System for Concrete Compliance and QA Risk Management

Re-establishing **Singapore operations for Ambient Systems** to lead the next wave of **AI + ESG transformation** in the built environment.

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"From delivery orders to ESG dashboards — making construction operations measurable, verifiable, and intelligently safe."

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